

Master IARFID
Reconocimiento de Escritura (RES)
Practical session
State-of-the-art HTR systems:Decoding

PRHLT-Group



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Description

- ▶ Deep Neural Networks are the state-of-the-art technology for handwritten text recognition.
- ▶ We will make a complete HTR system based on neural networks.
- ▶ First, we have trained an optical model based on Deep Neural Networks with Connectionist Temporal Classification (CTC).
- ▶ Secondly, we will:
 - Ⓐ Build a weighted finite state transducer to constrict the search space: TLG.fst
 - Ⓑ Test the TLG model
 - Ⓒ Get Word Graphs

B. Preparing the search graph

- ▶ A search graph is built with three WFSTs (T , L and G) compiled independently and combined as follows:

$$S = T \circ \min(\det(L \circ G))$$

T , L and G are the token, lexicon and grammar WFSTs respectively, whereas $\circ$, $\det$ and $\min$ denote composition, determination and minimization, respectively. The determination and minimization operations are needed to compress the search space, yielding a faster decoding.

- ▶ We shall need some scripts

```
wget --no-check-certificate \  
http://www.prhlt.upv.es/~mpastorg/RES/scripts.tgz  
  
tar xvzf scripts.tgz
```

B. Preparing the search graph: train the language model

④ Getting the corpus training

```
gawk -v TR_LIST=data/lists/train.lst \  
-v VAL_LIST=data/lists/val.lst '  
BEGIN{  
    while ((getline < TR_LIST) > 0) DICT[$1] = 1  
    while ((getline < VAL_LIST) > 0) DICT[$1] = 1  
}{  
    if ($1 in DICT) {  
        $1="";  
        print $0  
    }  
}' data/text/transcriptions.txt > data/text/train_words.txt
```

```
mkdir -p models/WFST
```

B. Preparing the search graph: L.fst

5 Getting the mapping from words to numbers

```
gawk '{for (w=2; w<=NF; w++){print $w}}' \
      data/text/transcriptions.txt | sort -u | \
gawk 'BEGIN{print "<eps> 0"; print "<s> "1; print "</s> 2"}
      {print $0, NR+2} END{ print "#0",NR+3; }' \
> models/WFST/wordsMap.txt
```

6 Build the lexicon

```
gawk 'BEGIN{
  print "<s> <space> #1"; print "</s> <space> #2";
}{
  N=split($1,CHARS,"");
  if ($1 !~ /<s>/ && $1 !~ /<\/s>/ &&
      $1 !~ /<eps>/ && $1 !~ /#0/){
    printf("%s ", $1);
    for(i=1; i<=N; i++)
      printf("%s ",CHARS[i]);
    printf("<space>\n")
  }
}' models/WFST/wordsMap.txt > models/WFST/lexicon.txt
```

B. Preparing the search graph: TLG model

① Getting the confidences matrix

```
HTRnn/get_confMats_binary.py --batch_size 24 \  
    --conf_matrix models/Optical/Conf_mat_test.ark \  
    models/Optical/BEST_MODEL data/data_test
```

② Build the T.fst, L.fst G.fst and the composition: TLG.fst

```
sed "s/~/<space>/" data/lists/symbols_train.lst > /tmp/symb  
mv /tmp/symb data/lists/symbols_train.lst  
  
./scripts/makeFST.sh 3 data/lists/symbols_train.lst \  
data/text/train_words.txt models/Optical/Conf_mat_test.ark
```

C. Testing the TLG model

- 1 mkdir results
- 2 Getting hypothesis using the TLG model

```
decode-faster --print-args=false \  
--beam=30.0 --max-active=5000 \  
--acoustic-scale=1.0 --allow-partial=true \  
--word_symbol-table=models/WFST/wordsMap.txt \  
models/WFST/TLG.fst ark:models/Optical/Conf_mat_test.ark \  
ark,t:results/test_TLG_numbers.hyp
```

- 3 Getting the tasas evaluating tool

```
wget --no-check-certificate \  
http://www.prhlt.upv.es/~mpastorg/RES/tasas.tgz  
  
tar xvzf tasas.tgz  
cd tasas  
gcc -O3 -o tasas tasas.c  
cd ..
```

C. Testing the TLG model

3 Preparing the test references

```
gawk -v TEST_LIST=data/lists/test.lst '
BEGIN{
    while ((getline < TEST_LIST) > 0) DICT[$1] = 1
}{
    if ($1 in DICT) {
        printf("%s ", $1)
        $1="";
        N=split($0, CHAR, "")
        for(i=1; i<=N; i++){
            printf("%s", CHAR[i])
        }
        printf ("\n")
    }
}' data/text/transcriptions.txt | \
    sed "s/\\s\\s*/ /g" > data/text/test_words.ref
```


C. Testing the TLG model

④ Getting WER and CER

```
scripts/int2sym.pl -f 2-200 models/WFST/wordsMap.txt \  
    results/test_TLG_numbers.hyp | \  
gawk -v TEST_LIST=data/text/test_words.ref 'BEGIN{  
    while ((getline < TEST_LIST) > 0) DICT[$1] = $0  
} {  
    if ($1 in DICT) {  
        N=split(DICT[$1],DIC)  
        for (i=2; i<NF; i++) printf("%s ", $i)  
        printf("%s#", $NF);  
        for (i=2; i<N; i++) printf("%s ", DIC[i])  
        printf("%s\n", DIC[N]);  
    }  
}' > results/test_TLG_hyp2ref  
  
echo -e "WER= "`tasas/tasas -f "#" -s " " results/test_TLG_hyp2ref`  
echo -e "CER= "`tasas/tasas -f "#" results/test_TLG_hyp2ref`
```

output -> CER = 2.66 WER = 8.99

D. Getting Word Graphs

① Get the word graphs

```
N_FEAT=`feat-to-dim ark:models/Optical/Conf_mat_test.ark -`  
./scripts/mkTopo.sh ${N_FEAT} > models/WFST/topology.hmm  
  
gmm-init-mono --print-args=false models/WFST/topology.hmm \  
${N_FEAT} models/WFST/model.hmm /tmp/auxTree  
  
mkdir results/lattices  
  
latgen-faster-mapped --beam=25.0 --max-active=5000 \  
--acoustic-scale=1.0 models/WFST/model.hmm \  
models/WFST/TLG.fst ark:models/Optical/Conf_mat_test.ark \  
"ark,t:|gzip -c > results/lattices/test-word-lat.gz" \  
ark,t:results/lattices/test-3gram-words-1bestLat.hyp
```

D. Getting Word Graphs

- 1 Label the edges with words.

```
zcat results/lattices/test-word-lat.gz | \  
perl scripts/int2sym.pl -f 3 models/WFST/wordsMap.txt \  
> results/lattices/test-word-lat
```

- 2 From Kali to HTK format.

```
mkdir -p results/lattices/words  
  
perl scripts/convert_slf.pl results/lattices/test-word-lat \  
results/lattices/words
```

D. Getting Word Graphs

1 Getting WER and CER of the 1_best on the word graphs.

```
scripts/int2sym.pl -f 2-200 models/WFST/wordsMap.txt \  
results/lattices/test-3gram-words-1bestLat.hyp | \  
gawk -v FILE_LIST=data/text/test_words.ref 'BEGIN{  
    while ((getline < FILE_LIST) > 0) DICT[$1] = $0  
    }{  
        if ($1 in DICT) {  
            N=split(DICT[$1],DIC)  
            for (i=2; i<NF; i++) printf("%s ", $i)  
            printf("%s#", $NF);  
            for (i=2; i<N; i++) printf("%s ", DIC[i])  
            printf("%s\n", DIC[N]);  
        }  
    }' > results/lattices/test_WG_hyp2ref  
  
echo -e "WER= "`tasas/tasas -f "#" -s " " \  
results/lattices/test_WG_hyp2ref`  
echo -e "CER= "`tasas/tasas -f "#" \  
results/lattices/test_WG_hyp2ref`
```

Getting word graphs for whole corpus

① Getting the confidences matrix for rest corpus

```
mkdir data/data_train_val
cp -r data/data_train/* data/data_val/* data/data_train_val/

HTRnn/get_confMats_binary.py --batch_size 24 \
    --conf_matrix models/Optical/Conf_mat_train_val.ark \
    models/Optical/BEST_MODEL data/data_train_val
```

D. Getting Word Graphs

① Get the word graphs

```
latgen-faster-mapped-parallel --num-threads=10 \  
--beam=25.0 --max-active=5000 \  
--acoustic-scale=1.0 models/WFST/model.hmm \  
models/WFST/TLG.fst ark:models/Optical/Conf_mats_train_val.ark \  
"ark,t:|gzip -c > results/lattices/train_val-word-lat.gz" \  
ark,t:results/lattices/train_val-3gram-words-1bestLat.hyp
```

② Label the edges with words.

```
zcat results/lattices/train_val-word-lat.gz| \  
perl scripts/int2sym.pl -f 3 models/WFST/wordsMap.txt \  
> results/lattices/train_val-word-lat
```

③ From Kali to HTK format.

```
perl scripts/convert_slf.pl results/lattices/train_val-word-lat \  
results/lattices/words
```

Extra work

Experiment without word language model but instead 8-gram character language model

8-grams of chars

① WFST for 8-gram chars

```
mkdir -p models/WFST
gawk 'BEGIN{
    cont=0;
    print "<eps>",cont++;
    print "<ctc>",cont++;
    print "<blk>",cont++; getline; getline;
}{ print $1,cont++; }
END{
    print "#0",cont++;
}' data/lists/symbols_train.lst > models/WFST/tokensMap.txt
```

```
scripts/ctc_token_fst.py models/WFST/tokensMap.txt | \
fstcompile --isymbols=models/WFST/tokensMap.txt \
--osymbols=models/WFST/tokensMap.txt | \
fstarcsort --sort_type=olabel > models/WFST/T.fst
```


8-grams of chars

① WFST for 8-gram chars

```
cat data/data_train/*.txt data/data_val/*.txt | \
awk '{
    gsub(" ", "~", $0);
    n=split($0, CHARS, "~");
    for (i=2; i<=n; i++)
        printf("%s ", CHARS[i]); printf("\n")
}' | sed "s/~/<space>/g" | ngram-count -text - \
-order 8 -wbdiscnt1 -kndiscnt \
-interpolate -lm models/LM/8gram-chars.arpa

arpa2fst models/LM/8gram-chars.arpa | fstprint | \
sed -e 's/<eps>/\#0/' -e 's/<s>/<eps>/g' -e 's/<\s>/<eps>/g' | \
fstcompile --isymbols=models/WFST/tokensMap.txt \
--osymbols=models/WFST/tokensMap.txt | fstrmepsilon | \
fstarcsort --sort_type=ilabel > models/WFST/G.fst

fsttablecompose models/WFST/T.fst models/WFST/G.fst \
models/WFST/TG.fst
```

8-grams of chars

① Getting the hmm model

```
N_FEAT=`feat-to-dim ark:models/Optical/Conf_mat_test.ark -`  
  
./scripts/mkTopo.sh ${N_FEAT} > models/WFST/topology.hmm  
  
gmm-init-mono --print-args=false models/WFST/topology.hmm \  
${N_FEAT} models/WFST/model.hmm /tmp/auxTree
```

② Getting hypothesis using the TG model

```
decode-faster --print-args=false --beam=30.0 --max-active=5000 \  
--acoustic-scale=1.0 --allow-partial=true \  
--word_symbol-table=models/WFST/tokensMap.txt models/WFST/TG.fst \  
ark:models/Optical/Conf_mat_test.ark \  
ark,t:./results/test_TG_numbers.hyp
```

C. Testing the TLG model

③ Preparing the test references

```
gawk -v TEST_LIST=data/lists/test.lst '
BEGIN{
    while ((getline < TEST_LIST) > 0) DICT[$1] = 1
}{
    if ($1 in DICT) {
        printf("%s ", $1)
        $1="";
        gsub(" ", "~", $0);
        N=split($0, CHAR, "~")
        for(i=1; i<=N; i++){
            printf("%s ", CHAR[i])
        }
        printf ("\n")
    }
}' data/text/transcriptions.txt | \
sed -e "s/~/<space>/g" -e "s/\\s\\s*/ /g" \
> data/text/test_chars.ref
```

C. Testing the TLG model

④ Getting WER and CER

```
scripts/int2sym.pl -f 2-200 models/WFST/tokensMap.txt \  
    results/test_TG_numbers.hyp | \  
gawk -v TEST_LIST=data/text/test_chars.ref 'BEGIN{  
    while ((getline < TEST_LIST) > 0) DICT[$1] = $0  
} {  
    if ($1 in DICT) {  
        N=split(DICT[$1],DIC)  
        for (i=2; i<NF; i++) printf("%s ", $i)  
        printf("%s#", $NF);  
        for (i=2; i<N; i++) printf("%s ", DIC[i])  
        printf("%s\n", DIC[N]);  
    }  
}' > results/test_TG_hyp2ref  
  
echo -e "WER= "`tasas/tasas -f "#" -s " " results/test_TG_hyp2ref`  
echo -e "CER= "`tasas/tasas -f "#" results/test_TG_hyp2ref`
```

output -> CER = 2.66 WER = 8.99

8-grams of chars

① Getting the word graphs

```
mkdir -p results/lattices8

latgen-faster-mapped --beam=25.0 --max-active=5000 \
  --acoustic-scale=1.0 models/WFST/model.hmm \
  models/WFST/TG.fst ark:models/Optical/Conf_mat_test.ark \
  "ark,t:|gzip -c > results/lattices8/test-word-lat.gz" \
  ark,t:results/test-8gram-chars-1bestLat.hyp
```

8-grams of chars

① Getting WER and CER of the 1_best on the word graphs

```
scripts/int2sym.pl -f 2-200 models/WFST/tokensMap.txt \  
results/lattices8/test-8gram-chars-1bestLat.hyp | \  
gawk -v FILE_LIST=data/text/test_chars.ref 'BEGIN{  
    while ((getline < FILE_LIST) > 0) DICT[$1] = $0  
}{  
    if ($1 in DICT) {  
        N=split(DICT[$1],DIC)  
        for (i=2; i<NF; i++) printf("%s ", $i)  
        printf("%s#", $NF);  
        for (i=2; i<N; i++) printf("%s ", DIC[i])  
        printf("%s\n", DIC[N]);  
    }  
}' > results/lattices/test_WG_hyp2ref  
  
echo -e "WER= "`tasas/tasas -f "#" -s " " \  
results/lattices/test_WG_hyp2ref`  
echo -e "CER= "`tasas/tasas -f "#" \  
results/lattices/test_WG_hyp2ref`
```